Temi Otun

otun@ualberta.ca | GitHub | LinkedIn | Website

Education

University of Alberta Edmonton, AB

Bachelor of Science, Computing Science – Artificial Intelligence Specialization, Minor in Business *Expected Apr. 2027*

- **Awards:** Jason Lang Scholarship, Dean's List

Experience

Data Engineering Intern May 2026 – Aug. 2026

Royal Bank of Canada (RBC) – Amplify Program *Vancouver, BC*

- Will collaborate with a cross-functional student team to design and prototype a data-driven solution addressing a real RBC business challenge through rapid iteration and experimentation
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Machine Learning Intern Jan. 2025 – Aug. 2025

The Metabolomics Innovation Centre *Edmonton, AB*

- Enhanced long-range temperature forecasting system, resulting in up to **47%** lower MAE with an average **24.3%** improvement across a six-year test set compared to previous approaches
- Developed multivariate time series models for precipitation forecasting, incorporating **100+** climate features and outperforming baseline models in **73%** of test years
- Assembled forecasting pipelines with advanced feature engineering and benchmarked **20+** ML models, attaining a **42%** accuracy improvement over baseline methods
- Automated SQL pipelines to ingest weather and climate data from APIs and web scrapers, generating new feature combinations to support testing and improvement of forecasting models

Machine Learning Research Assistant Sept. 2024 – Apr. 2026

University of Alberta *Edmonton, AB*

- Contributed to **3** machine learning systems in computational psychiatry and predictive healthcare, including dementia detection and ECG signal modeling on large-scale clinical datasets
- Synthesized insights from **40+** research seminars on survival analysis and disease prediction, applying advanced statistical and ML methods to strengthen ongoing projects
- Documented and analyzed **15+** ML experiments, applying feature engineering, hyperparameter tuning, and evaluation pipelines to improve performance across classification and regression tasks

Data Management Intern Jan. 2024 – May 2024

InfoStrux *Vancouver, BC*

- Optimized **25+** SQL queries in Snowflake, reducing execution time by up to **60%** and boosting performance of business intelligence dashboards
- Partnered with a senior data engineer to design Snowflake staging and curated layers for **3** datasets; wrote and tuned **30+** queries, improving data reusability and cutting time-to-insight by **40%**
- Constructed and maintained **10** database schemas in Snowflake to support diverse data types, improving pipeline efficiency and data flow

Projects

Emotion Detection CNN | *PyTorch, ML, Computer Vision, AI, Data Processing, Python* GitHub

- Trained a deep convolutional neural network in PyTorch on the FER-2013 dataset with **32K+** labeled images for multi-class facial emotion recognition
- Achieved **70%** test accuracy across **7** emotion classes and integrated webcam inference for real-time emotion detection
- Applied augmentation, batch normalization, and ReLU activations to improve model performance

Lung Cancer Detection | *AI, Python, Machine Learning, AI engineering* GitHub

- Achieved **99%** recall, **94%** accuracy, **95%** precision, and **97%** F1 score on the best classification model
- Built and compared multiple ML models, including SVM, k-NN, and Random Forest, to determine the most effective classification approach

Technical Skills

Programming: C, Java, Python, SQL
Libraries & Frameworks: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, OpenCV, React
Tools & Platforms: Git, Linux, Bitbucket, Docker, Snowflake, Kubernetes, AWS S3